Course: [BCS-302] IT Infrastructure Management

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: LECTURE NOTES**

Module / Unit: Unit 1 **Faculty Author: Prof. Amit Saxena (Dept. of C**

Unit 1 Lecture Notes: Modern Data Center Infrastructure & Virtualization

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1. Enterprise IT Infrastructure Architecture

- Tiered Data Center Classification (Tier I to IV uptime and redundancy benchmarks).
- Server Hardware Architecture: Rack, blade, and hyper-converged compute nodes.
- Storage Topologies: DAS (Direct Attached), NAS (Network Attached), SAN (Storage Area Network).

2. Server Virtualization & Cloud Foundations

- Hypervisor Typologies: Type-1 (Bare Metal / ESXi, KVM) vs Type-2 (Hosted / VirtualBox).
- Resource Pooling: vCPU scheduling, NUMA node topology, memory overcommit techniques.
- High Availability (HA) & Disaster Recovery (DR): RTO and RPO metrics.